Contents

1 Gauge-Constrained Spectral Geometry

1.1 Unified Master Theory of the SU(N) Wilson / Kogut–Susskind Spectral Program

Audit-hardened release: v1.4 (2026-08-08).

Master synthesis date: 2026-08-08 · **v1.4:** retains v1.3 but incorporates the independent `SIMULATIONS/` audit as a proof-status authority for the $SU(3)$ strong-coupling tower. The hand-checkable $O(y^2)$ homological factorization and the structural $O(y^3)$ triality mechanism remain strong. The $O(y^4)$ rational kernel/pencil is retained as an exact-arithmetic computational ledger, but its end-to-end theorem status is explicitly withheld until the shared effective-Hamiltonian, normalization, independent-regression, interval-enclosure, and near-Γ nonuniformity gates are closed. No coefficient is deleted merely because the certificate chain is incomplete. **Supersedes as a *narrative* unification (not as a status authority):** the seven canonical ledgers of 2026-07-13, the updated flat-band theorem record of 2026-07-14, the strengthened GCSG synthesis of 2026-07-13, and the 2026-07/08 frontier notes (Hodge shape space; singularity atlas; certified exceptional points; crossover v1; Wilson–Bergman weight theorem; rank-1/2 Bergman transfer; tight-connections review). **Status authorities remain the canonical ledgers.** Where ledgers conflict, this document applies the third-pass adjudication rule: the latest scope-specific combined ledger controls its declared domain; elsewhere the narrowest supported statement is used. **Status vocabulary: Proven · Exact computational ledger — audit pending · Computationally verified · Conditional · Rejected · Conjectural · Open**. A computational ledger is never promoted to a theorem solely because its arithmetic is exact; the derivation and certificate dependencies must also be closed.

1.1.1 v1.4 consolidation and authority rule

This document separates three regimes that use related algebra but different expansion parameters:

1. **compact local class stiffness:** $\beta \to \infty$ for the one-plaquette class Hamiltonian;
2. **strong-coupling spatial mobility:** $u = \beta_H/6 \to 0$ for the one-flux band;
3. **global Wilson-measure stability:** finite/infinite-volume probabilistic control.

No coefficient is transferred between these regimes without an explicit bridge. The fourth-order shape-space theorem is an ambient symmetry classification. For Layer II, `PAPER homological flat bands.md` remains the rank/status authority relative to its accepted certificates, while `Pasted markdown.md` (the independent

`SIMULATIONS/` audit assembled 2026-08-08) controls proof-status claims about the $SU(3)$ $O(y^4)$ computational tower. The detailed $SU(4)$ theorem/certificate remains the narrow authority for its exceptional sector. When a coefficient ledger and an audit disagree only about proof closure, the numerical ledger is retained and the proof status is downgraded rather than silently discarded.

1.1.2 v1.3 change log — 2026-08-08

The 2026-08-08 homological-flat-band paper changes the Layer-II registry in four ways:

1. $SU(5)$ and $SU(6)$ are no longer listed as open at the axial fourth-order level.
2. The exact axial coefficient is universal for every $N \geq 3$,

$$\alpha_N^{\mathrm{pen}} = \frac{640}{N(N^2-1)^3}.$$

3. The physical tier-collapse statement $B_N^{\mathrm{shp}} = D_N^{\mathrm{shp}} = 0$ now spans the complete rank partition $N \geq 3$ covered by the accepted certificates/stable theorem.
4. $SU(3)$ is exceptional in the second pencil coefficient:

$$\Delta\beta_3^{\mathrm{pen}} = -\frac{25}{64},$$

so quotient-scalarity fails at $N = 3$ and must not be promoted to an all-rank theorem.

The compact local-class theory, seam theory, transfer/certificate layer, and projected-capacity no-go/firewall retain their v1.2 scope unless explicitly changed below.

1.1.3 v1.4 audit-hardening change log — 2026-08-08

The independent `SIMULATIONS/` re-audit does **not** overturn the central $SU(3)$ fourth-order rational ledger, but it changes the authority level of the full $O(y^4)$ theorem chain.

1. The $O(y^2)$ incidence identity

$$S(k) + 4I = B(k)B(k)^\dagger$$

and the all-orders annihilation criterion $BMB^\dagger|_{\ker B^\dagger} = 0$ are promoted as the cleanest hand-checkable Layer-II theorem package.
2. The $O(y^3)$ bare-link/triality argument remains structural; the numerical coefficient is exact arithmetic but still benefits from the requested cold replay.
3. The $SU(3)$ $O(y^4)$ coefficient values, 189-record kernel, bandwidth, and ray curvatures are retained as an **exact computational ledger**. Their end-to-end theorem status is pending closure of six load-bearing gates: $PVP = aP$, degenerate-space folded-effective-Hamiltonian regression, full Stage-3H independent regression, magnetic-prefactor normalization, outward-rounded interval certification, and the nonuniform $k \to \Gamma$ touching analysis.
4. The global interval certificate is therefore described as **numerically decisive but not formally interval-rigorous as currently coded**; the identified leaks are mechanical to fix.
5. The master no longer asserts that the Aug. 8 simulation variable can be relabeled to canonical $u = \beta_H/6$ without numerical rescaling. That bridge is conditional until the magnetic-prefactor gate is closed.
6. The rejected Schur/Haar-Hessian "Theorem B" and the unimplemented 4D $SU(2)$ θ/TRG branch are explicitly quarantined from the master theorem dependency graph. The analytical Wilson+Haar Hessian formula itself remains salvageable as a separate technical result.
7. Strong-coupling extrapolation to the continuum remains **consistent, not controlled**; the finite series must not be advertised as a convergent continuum prediction.

1.2 0. The theory in one statement

Gauge-constrained spectral geometry is a five-layer theory of spectra in compact-gauge lattice systems:

1. **Internal spectral stiffness** — the geometry of conjugacy classes and the Weyl measure fixes local gaps and their exact asymptotic coefficients (the one-plaquette layer).
2. **Homological mobility** — the position of an excitation inside the cellular chain complex of space fixes when, and at what perturbative order, it can move (the flat-band / caging layer).
3. **Seam analyticity** — a finite atlas of complex exceptional points governs the crossover between strong- and weak-coupling towers, and symmetric functions of colliding branches are the analytic continuers (the singularity layer).
4. **Transfer geometry and certificate duality** — the correct norm for tails and transfers is a Wilson–Bergman weighted norm on the Weyl alcove; positivity certificates live in a two-cone (pointwise x character) geometry whose large-N contraction is obstructed by sign-uncertainty (the certificate layer).
5. **Marginal stability** — exact marginalization (Helffer–Sjöstrand leakage) and rooted incidence-shadow capacity determine how much spectral protection survives coupling to unresolved degrees of freedom (the global layer).

The unification is **structural, not asymptotic**: the three coupling regimes (strong-coupling u-expansion, compact large-β well, global Wilson measure) are governed by the same operators and filtrations but are never conflated into one expansion. This regime firewall is itself part of the theory.

1.3 1. Objects and canonical conventions

- **Spatial complex.** $T_L^d = (\mathbb{Z}/L\mathbb{Z})^d$ with links, plaquettes, cubes; boundary maps ∂_2, ∂_3; coboundaries d_1, d_2; $D = d_1$.
- **Hamiltonian.** Kogut–Susskind $H = \frac{g^2}{2a} \sum E^2 + \frac{2N}{ag^2} \sum_p P_p$; one-plaquette class reduction $H = \frac{1}{2} C_2 + \beta V$ on class functions of $SU(N)$.
- **Strong-coupling normalization firewall.** The canonical Hamiltonian ledger uses $\boxed{u = \beta_H/6}$. The historical definition $y = 2\beta_H/3$ remains **Rejected**. However, the independent `SIMULATIONS/` audit found that stages 3F/3H explicitly exclude a convention-dependent magnetic-operator prefactor that is not later closed by a gate. Therefore the Aug. 8 simulation-tower coefficients are kept in their **source normalization** until that prefactor is proved. Agreement with the canonical u-ledger is strong consistency evidence, but v1.4 no longer treats "replace `y` by `u` with no rescaling" as a proved bridge.
- **Weyl reduction.** After Weyl-denominator conjugation the class kinetic operator is the A_{N-1} trigonometric Calogero–Sutherland operator at the free-fermion point $\kappa = 1$ (**Proven**, imported equivalence). Radial Laguerre parameter of the Weyl-Gaussian class sector: $\boxed{\alpha_N = (N^2 - 3)/2}$ (**Proven**).
- **Incidence cage.** For any coefficient of the factorized form $H^{(n)} = a_n I + B M_n B^\dagger$, the sector $\ker B^\dagger$ is shifted rigidly (**Proven**). Consistently oriented cube boundaries lie in this kernel because $\partial_2 \partial_3 = 0$.
- **Status governance.** Precedence when sources conflict: exact self-contained derivation > independent direct computation > internally consistent saved output > later consistent version > stale diagnostics. A filename saying "FINAL" never overrides a failed invariant check.

1.4 2. The five structural laws

These are the load-bearing organizing principles; every theorem in §§3–7 instantiates at least one.

Law 1 (Filtration–Rigidity–Escape). Identify an exactly preserved filtration; prove rigidity inside it; compute the first escaped coefficient exactly. *Instances:* internally, the $SU(3)$ even sector stays inside the radial invariant algebra $\mathbb{R}[p_2]$ until p_3^2 escapes at order $\beta^{-1/2}$ with exact shift $\sqrt{6}/576$; spatially, the one-flux branch stays inside the incidence ideal through $O(u^3)$ and escapes at $O(u^4)$ with exact coefficients

(q, A, B); at the seam, the raw gap escapes analyticity at the vacuum EP while its square remains inside the symmetric-function algebra of the colliding pair.

Law 2 (Kernel–Resolvent Duality). One incidence operator plays two spectral roles: $\ker d_1^*$ hosts protected closed-surface carriers, while $(m^2 I + \alpha d_1^* d_1)^{-1}$ controls exponentially localized propagation in the complementary link sector. Both faces are exact on the periodic cubic complex (**Proven**):

$$(m^2 I + \alpha D^* D)^{-1}_{\mu\nu}(p) = \frac{\delta_{\mu\nu} + (\alpha/m^2) h_\mu(p)\overline{h_\nu(p)}}{m^2 + \alpha \hat{p}^2}, \qquad h_\mu(p) = e^{ip_\mu} - 1.$$

Law 3 (Two-Cone Certificate Duality and its contraction). On class functions, multiplication by h is PSD iff $h \geq 0$ pointwise; convolution by h is PSD iff all character coefficients $\hat{h}_R \geq 0$. Gap certificates decompose against exactly these two cones plus squares. Under 't Hooft-scaled weak coupling the compact cone contracts to a Euclidean cone of effective dimension $\sim N^2/2$ (the GUE identification with radial weight $u^{\alpha_N} e^{-u}$), where Mellin-strip sign-uncertainty bounds how tightly any fixed-degree certificate can localize — the structural mechanism behind the ledger's declared fixed-rank non-uniformity in N. (Framework **Proven/imported**; the specific limitation transplant is a pre-registered program, §12.)

Law 4 (Seam Analyticity). The convergence of every coupling-expansion in the program is governed by a finite, certifiable atlas of complex exceptional points of the complex-symmetric pencil $H(\beta)$. Symmetric functions of a colliding pair are single-valued through the collision; therefore the correct global objects are squared gaps and pair-symmetric combinations, not raw gaps.

Law 5 (Rooted Marginal Stability). Global fixed-window "firewalls" are false (Bernoulli no-go, **Proven**); the correct stability object is a rooted defect-animal expansion in which the incidence-shadow capacity $\mathcal{K} \leq 1$ contributes exactly one exponential mark $e^{s\gamma}$. Exact marginalization obeys the Helffer–Sjöstrand leakage bound $\nabla_x^2 S_{\text{eff}} \succeq \mathbb{E}\nabla_{xx}^2 S - M^2 I/\gamma$ (**Proven**).

1.5 3. Layer I — Internal spectral stiffness (one-plaquette theorems)

1.5.1 3.1 The compact SU(3) class-gap theorem — Proven (with $O(\beta^{-1})$ remainder)

$$\boxed{\Delta^+_{\mathrm{SU}(3)}(\beta) = \sqrt{\tfrac{2\beta}{3}} - \tfrac{5}{16} - \tfrac{311\sqrt{6}}{9216}\,\beta^{-1/2} + O(\beta^{-1})} \qquad \boxed{c_1 - c_1^{\text{radial}} = \tfrac{\sqrt{6}}{576}}$$

with $H_1 = -p_2^2/96$, $H_2 = \sqrt{6}\,(p_2^3/11520 + p_3^2/8640)$; non-radial multiplicity $\chi_{\text{nr}} = 2$. The first even escape from $\mathbb{R}[p_2]$ is through p_3^2. Independently confirmed by a converged compact Peter–Weyl execution ($K = 80$, $\beta = 50 \ldots 3200$) (**Computationally verified**).

Remainder closure now in-corpus (v1.1). The analytic $O(\beta^{-1})$ step is proved, not imported: exact Weyl conjugation $J(\frac{1}{2}C_2)J^{-1} = \frac{1}{4}(-\Delta_{\mathfrak{t}} - |\rho|^2)$ reduces H_β to a flat torus Schrödinger operator with **no untracked Haar/metric terms**; the Wilson potential $W = 1 - \frac{1}{3}\sum_j \cos\theta_j \geq 0$ has the identity as its unique nondegenerate minimum ($D^2W(0) = \frac{1}{3}I_2$), supplying the confinement hypotheses; with $h = \beta^{-1/2}$ and well scaling $a = (3/2)^{1/4}$ the rescaled operator has the exact normal form $H_0 + hH_1 + h^2H_2 + h^3R_h$ with $H_0 = \frac{1}{2\sqrt{6}}(-\Delta_z + p_2)$; an **equivariant harmonic-well lemma** (Agmon/IMS localization, symmetry-restricted polynomial-Gaussian quasimodes with residual $O(h^4)$; a special case of the Charles–Vũ Ngọc semiclassical Birkhoff normal form) then identifies the two simple even levels (ψ_0 =const, $\psi_1 \propto p_2 - 4$) and delivers the three-term gap with $O(\beta^{-1})$ remainder. The scalar $-|\rho|^2/4 = -\frac{1}{2}$ cancels from the gap. Exact ingredients: $\Delta_{H_2} = 19\sqrt{6}/576$, $\Delta_{\text{res}} = -205\sqrt{6}/3072$, moment table $\langle p_2^3\rangle, \langle p_3^2\rangle = (120, 5)_{\psi_0}, (480, 20)_{\psi_1}$.

1.5.2 3.2 Fixed-rank SU(N) coefficients

- Leading C-even gap $\sqrt{2\beta/N}$; $c_0^{(N)} = -(2N^2-3)/(16N)$ — **Proven** for fixed $N \geq 3$.
- Lowest C-odd shell is $s = 3$; leading C-odd gap $\sqrt{9\beta/(2N)}$; $c_0^{(N),-} = -3(N^2-3)/(16N)$ — **Proven**, $N \geq 3$.

- Candidate $\beta^{-1/2}$ closed forms

$$c_1^{(N),+} = -\frac{\sqrt{2}\,(6N^4-24N^2+41)}{1024\,N^{3/2}}, \qquad c_1^{(N),-} = -\frac{\sqrt{2}\,(14N^4-97N^2+290)}{1536\,N^{3/2}}$$

— **Proven at** $N=3$ (even), **Computationally verified (exact arithmetic) for** $N=3,\dots,12$, **Conjectural** for unrestricted fixed N pending a symbolic-N Gram transcript.

1.5.3 3.2b The β^{-1} order — exact third-order coefficients (new, v1.1)

Third-order Rayleigh–Schrödinger on the Weyl-Gaussian shells, with the exact operator stack $H_1=-P_4/48$, $H_2=\sqrt{2N}\,P_6/2880$, $H_3=-N\,P_8/161280$ and shell energies $E_s^{(0)}=s/\sqrt{2N}$, yields closed forms for the next coefficient in both sectors:

$$\boxed{c_2^{(N),+} = -\frac{60N^6-401N^4+1522N^2-2297}{49152\,N^2}, \qquad c_2^{(N),-} = -\frac{95N^6-981N^4+5853N^2-15335}{49152\,N^2},}$$

with the resolvent part of c_1 separately closed: $q_{\text{res}}^{(N)}=-(34N^4-120N^2+171)/(3072N^2)$. **Verification (exact rational arithmetic):** even for $N=6,\dots,12$; odd for $N=7,\dots,13$; and — resolving the power-sum overcompleteness that blocks small ranks — **exactly at** $N=3$ in the true rank-two invariant basis $\{p_2,p_3\}$ with the SU(3) reductions $P_4=p_2^2/2$, $P_6=p_2^3/4+p_3^2/3$, $P_8=p_2^4/8+4p_2p_3^2/9$:

$$\boxed{c_2^+(3)=-\tfrac{5665}{110592}, \qquad c_1^-(3)=-\tfrac{551\sqrt{6}}{13824}, \qquad c_2^-(3)=-\tfrac{53}{864}.}$$

The SU(3) local class gaps are therefore known **four terms deep in both C-sectors**:

$$\Delta^+(\beta)=\sqrt{\tfrac{2\beta}{3}}-\tfrac{5}{16}-\tfrac{311\sqrt{6}}{9216}\beta^{-1/2}-\tfrac{5665}{110592}\beta^{-1}+\dots,$$

$$\Delta^-(\beta)=\sqrt{\tfrac{3\beta}{2}}-\tfrac{3}{8}-\tfrac{551\sqrt{6}}{13824}\beta^{-1/2}-\tfrac{53}{864}\beta^{-1}+\dots$$

An independent odd-sector compact Peter–Weyl audit (irrep basis, C-splitting under $(p,q)\leftrightarrow(q,p)$, $K=80$, $\beta\le 3200$) confirms the odd three-term law, and its β^{-1} fit selects $-53/864$ over the superseded direct candidate $-1781/55296$; the exact-arithmetic identity is the authority. **Status:** the closed forms are **Computationally verified (exact arithmetic)** at the listed ranks and at $N=3$, **Conjectural** as unrestricted fixed-N formulas (open verification gaps: even $N=4,5$; odd $N=4,5,6$); the analytic $O(\beta^{-3/2})$ remainder at this order awaits a one-order extension of the harmonic-well lemma of §3.1, which as written closes $O(\beta^{-1})$. This discharges local-frontier priority 4 ("next $O(\beta^{-1})$ coefficients") at the certificate level.

1.5.4 3.3 Polarity-excess law — Proven corollary of 3.2

The C-odd and C-even leading terms cancel in the ratio 3/2, leaving an exact rank-diagnostic constant:

$$\boxed{\Delta_-^{(N)}-\tfrac{3}{2}\,\Delta_+^{(N)}=\frac{9}{32N}+O(\beta^{-1/2}).}$$

(The $\beta^{-1/2}$ refinement inherits the Conjectural status of the c_1 candidates.)

1.5.5 3.4 The odd-Casimir staircase and cubic lock — Proven

With $U=e^{iX}$, $P_k=\operatorname{Tr}X^k$:

$$SU(3):\ \operatorname{ImTr}U=-\frac{P_3}{6}\prod_{j=1}^{3}\frac{\sin(\theta_j/2)}{\theta_j/2}, \qquad SU(4):\ \operatorname{ImTr}U=-\frac{P_3}{6}\prod_{\{ij\}\subset\{123\}}\frac{\sin((\theta_i+\theta_j)/2)}{(\theta_i+\theta_j)/2},$$

both form factors positive on the alcove — $SU(3)$ and $SU(4)$ are **exactly cubic-locked** ($\operatorname{ImTr}U$ shares sign and interior nodal set with $-P_3$). The lock breaks at $SU(5)$: $P_5=\frac{5}{6}P_2P_3+5e_5$ makes e_5 an independent

C-odd direction for $N \geq 5$ (explicit counterexample with $P_1 = P_3 = 0,\ P_5 \neq 0$). Primitive odd directions: e_3 (all $N \geq 3$); $+e_5$ at $N \geq 5$; $+e_7$ at $N \geq 7$; … The exact Gram calculations independently exhibit the new degree-5 channel at $N = 5$ and degree-7 at $N = 7$. **Derived lattice operators:** improved cubic source $\mathcal{O}_3^{\mathrm{imp}} = (32A - B)/24 = -P_3/6 + P_7/1260 + O(|X|^9)$ (quintic contamination cancelled) and primitive rank-five source $\mathcal{O}_5^{\mathrm{prim}} = B - 8A + 2EA = e_5 + O(|X|^7)$, with $A = \mathrm{Im}\,\mathrm{Tr}\,U$, $B = \mathrm{Im}\,\mathrm{Tr}\,U^2$, $E = N - \mathrm{Re}\,\mathrm{Tr}\,U$. For $SU(5)$, $e_5(X) = \det X$.

1.5.6 3.5 Leakage diagnostics and the radial-tail no-go

Exact four-channel leakage quartic $\lambda^4 - \frac{215}{768}\lambda^2 - \frac{175}{13824}\lambda + \frac{25}{294912}$ (**Proven**); Perron root $\rho_3 = 0.5501615335\ldots$ (**Computationally verified**, finite-channel only). The one-resolvent/Schur-symmetric radial-tail obstruction is **Proven**: Laguerre quartic off-diagonals grow $\sim n^2$ against the present denominator family, so finite leakage numbers cannot be promoted to a full-channel constant by that route. Compressed compact Peter–Weyl high-block inverse is $O(R^{-2})$ (**Proven**); the buffered full-resolvent/transfer theorem in compact shells is **Open** (SU3-10).

1.6 4. Layer II — Homological mobility (the one-flux C-odd band)

1.6.1 4.1 Exact second-order rank/topology factorization — Proven

In the one-excitation sector, orient each plaquette by its cellular boundary and let

$$s(p, p') = \epsilon_p(\ell)\epsilon_{p'}(\ell)$$

for adjacent plaquettes sharing the link ℓ. The fixed-rank representation-theory contraction reduces the entire C-odd second-order hopping problem to one positive scalar,

$$\boxed{t_N = \frac{2N(N^2-4)}{(N^2-1)(2N^2-1)(4N^2-9)}, \qquad N \geq 3.}$$

Thus representation theory supplies the amplitude while the cellular boundary operator supplies the band geometry.

The rank-cubic limit is

$$0 < N^3 t_N < \frac{1}{4}, \qquad N^3 t_N \nearrow \frac{1}{4},$$

with exact deficit

$$\boxed{\frac{1}{4} - N^3 t_N = \frac{2N^4 + 31N^2 - 9}{4(N^2-1)(2N^2-1)(4N^2-9)}.}$$

1.6.2 4.2 Incidence factorization, complete Bloch spectrum, and Hodge self-duality — Proven

Set $u_j = 1 - e^{ik_j}$ and

$$\widetilde{N}(k) = \begin{pmatrix} u_2 & -u_1 & 0 \\ u_3 & 0 & -u_1 \\ 0 & u_3 & -u_2 \end{pmatrix}.$$

The signed face adjacency satisfies

$$\boxed{S(k) + 4I = \widetilde{N}(k)\widetilde{N}(k)^\dagger}.$$

Writing

$$q(k) = \sum_j |u_j|^2 = 4\sum_{j=1}^{3} \sin^2 \frac{k_j}{2},$$

one obtains the complete spectrum

$$\boxed{\operatorname{spec} S(k) = \{-4, -4 + q(k), -4 + q(k)\}.}$$

The lowest C-odd branch is therefore exactly dispersionless at this order for every $N \geq 3$.

In centered gauge, with $s_j = \sin(k_j/2)$,

$$M(k) = 2 \begin{pmatrix} s_2 & -s_1 & 0 \\ s_3 & 0 & -s_1 \\ 0 & s_3 & -s_2 \end{pmatrix},$$

and for

$$J(z_{12}, z_{13}, z_{23}) = (z_{23}, -z_{13}, z_{12})$$

one has

$$\boxed{JM = -2[s]_\times}, \qquad \boxed{M^\dagger M = JMM^\dagger J^T = 4(|s|^2 I - ss^T)}.$$

The face and link Hodge Hamiltonians are therefore gauge-Hodge equivalent to the same transverse lattice-Maxwell operator. Their kernels are the same longitudinal line after Hodge rotation,

$$\ker M = \operatorname{span}\{s\}, \qquad J \ker M^\dagger = \operatorname{span}\{s\}.$$

On a sphere around Γ, $\hat{s} = s/|s|$ gives a unit embedded map $S^2 \to \mathbb{RP}^2$. Since adding one trivial real orbital changes the target to $\mathbb{RP}^3$ with $\pi_2(\mathbb{RP}^3) = 0$, the correct topology is a

$$\boxed{\text{fragile unit Hodge hedgehog, not stable BDI protection.}}$$

1.6.3 4.3 Exact torus degeneracy and finite-volume isolation — Proven

The flat carrier is the two-cycle space:

$$0 \to \operatorname{im} \partial_3 \to \ker \partial_2 \to H_2(T^3; \mathbb{C}) \to 0.$$

Hence on an L^3 torus,

$$\boxed{\dim \ker \partial_2 = (L^3 - 1) + 3 = L^3 + 2.}$$

After removing the complete flat eigenspace, the first adjacency level is

$$\boxed{4 \sin^2 \frac{\pi}{L}}.$$

The isolation scale is therefore L^{-2}, not L^{-1}.

A falsifiable integer implementation check is: $L^3 + 2$ flat states on T^3, $L^3 + 1$ on $T^2 \times I$, and L^3 on an open box.

1.6.4 4.4 Rank-cubic mobility and third-order rigidity

The complete second-order C-odd one-plaquette manifold has width

$$W_N^-(u) = 12 t_N u^2 + O(u^3),$$

so

$$W_N^-(u) \sim \frac{3u^2}{N^3}.$$

This is the exact rank-cubic mobility suppression.

For the $SU(3)$ caged branch, the source-certified expansion is

$$m^{SU(3)}_{-,\mathrm{flat}}(u) = \frac{8}{3} + u + \frac{11}{306}u^2 - \frac{109151}{249696}u^3 + O(u^4).$$

The 2026-08-08 paper explains the third-order rigidity by the bare-link/tromino mechanism: every three-distinct-plaquette numerator vanishes at $O(u^3)$, so the third-order effective Hamiltonian retains the second-order incidence structure. The coefficient itself remains flagged for a clean cold rerun before submission, as stated in that paper.

1.6.5 4.5 Generic fourth-order obstruction space — Proven (exact orbit enumeration)

Let

$$a_i = 4\sin^2\frac{k_i}{2}, \qquad q = a_1 + a_2 + a_3,$$
$$e_2 = a_1a_2 + a_1a_3 + a_2a_3, \qquad e_3 = a_1a_2a_3.$$

The most general nonconstant cubic-invariant fourth-order correction on the generic flat fiber is

$$\boxed{\varepsilon_4(k) = c_0 + A^{\mathrm{shp}}_N q + B^{\mathrm{shp}}_N e_2 + C^{\mathrm{shp}}_N \frac{4e_2}{q} + D^{\mathrm{shp}}_N \frac{e_3}{q}.}$$

The obstruction space splits into two infrared tiers,

$$\boxed{\mathcal{Q}_4 = \underbrace{\mathrm{span}\{q, 4e_2/q\}}_{L^{-2}} \oplus \underbrace{\mathrm{span}\{e_2, e_3/q\}}_{L^{-4}}.}$$

With checkpoints $X = (\pi, 0, 0)$, $M = (\pi, \pi, 0)$, $P = (\pi, \pi/2, 0)$, and $R = (\pi, \pi, \pi)$, define $\Delta_K = \varepsilon_4(K) - \varepsilon_4(\Gamma)$. Then

$$A = \frac{\Delta X}{4}, \qquad B = \frac{\Delta X + 4\Delta M - 6\Delta P}{16},$$
$$C = \frac{3(2\Delta P - \Delta M - \Delta X)}{8}, \qquad D = \frac{3(\Delta R - 6\Delta M + 6\Delta P)}{16}.$$

The identities

$$6\Delta P - 4\Delta M - \Delta X = -16B,$$
$$\Delta R - 2\Delta M + \Delta X = 16B + \frac{16}{3}D$$

are algebraic consequences of the basis and are not independent validation gates.

1.6.6 4.6 Physical tier collapse — Exact within the accepted fourth-order ledgers; shared proof gates pending

At every rank now resolved,

$$\boxed{B^{\mathrm{shp}}_N = D^{\mathrm{shp}}_N = 0.}$$

Thus the physical fourth-order correction lies entirely in

$$\boxed{\mathrm{span}\left\{q, \frac{4e_2}{q}\right\}.}$$

The generic L^{-4} obstruction tier is allowed by cubic symmetry but is not selected by the exact microscopic contraction. This is a dynamical selection rule, not a symmetry identity.

In the older two-invariant pencil notation,

$$c_{4,N}(k) = q_N + \frac{\alpha^{\mathrm{pen}}_N \sum_i X_i^2 + \beta^{\mathrm{pen}}_N \sum_{i<j} X_iX_j}{2\sum_i X_i}, \qquad X_i = 1 - \cos k_i,$$

the conversion is

$$\boxed{A_N^{\rm shp} = \frac{\alpha_N^{\rm pen}}{4}}, \qquad \boxed{B_N^{\rm shp} = 0},$$

$$\boxed{C_N^{\rm shp} = \frac{\beta_N^{\rm pen} - 2\alpha_N^{\rm pen}}{16}}, \qquad \boxed{D_N^{\rm shp} = 0}.$$

1.6.7 4.7 Universal axial mobility law — Exact certificate-backed rank identity; audit-hardened proof status

For every integer $N \geq 3$,

$$\boxed{\alpha_N^{\rm pen} = \frac{640}{N(N^2-1)^3}}.$$

The accepted coefficient registry is finite and case-complete: the stable-rank walled-Brauer contraction covers every $N \geq 7$, while finite-rank certificates give

$$\alpha_3^{\rm pen} = \frac{5}{12}, \qquad \alpha_4^{\rm pen} = \frac{32}{675}, \qquad \alpha_5^{\rm pen} = \frac{1}{108}, \qquad \alpha_6^{\rm pen} = \frac{64}{25725}.$$

Direct substitution verifies the same rational law at all four exceptional ranks. In v1.4 this is an exact cross-rank coefficient identity **relative to the accepted rank certificates**. If those rank certificates share the same unclosed degenerate folded-effective-Hamiltonian or magnetic-normalization step identified in the `SIMULATIONS/` audit, their final theorem-grade promotion inherits that dependency; the rank identity itself is not numerically altered by the audit.

The threshold $N \geq 7$ has a structural origin: a fourth-order word contains at most six character factors, so a determinant/N-ality channel requires $|p-q| = N$ with $p+q \leq 6$, impossible for $N \geq 7$. Hence the exceptional set is exactly

$$\boxed{\{3,4,5,6\}}.$$

The first fourth-order finite-volume lift of the caged branch is

$$\boxed{\Delta_{N,L}^{(4)} = \alpha_N^{\rm pen} \sin^2 \frac{\pi}{L}},$$

independent of the second pencil coefficient.

1.6.8 4.8 Exceptional-rank ledger and the $SU(3)$ anomaly

The resolved determinant sectors leave the universal axial coefficient unchanged:

Rank	exceptional families	certified rest correction	$\Delta\alpha_N^{\rm pen}$
$SU(4)$	$(4,0),(0,4),(5,1),(1,5)$	$-304746539168/160249753125$	0
$SU(5)$	none	0	0
$SU(6)$	$(6,0),(0,6)$	$6/343$	0

For $SU(5)$ the mod-5 scan is empty across the recorded 895,524 support/output pairs. For $SU(6)$ the determinant channel is on-site and produces $\Delta q_6 = 6/343$.

The second pencil coefficient has one exceptional anomaly. The stable expression agrees with the independently certified values at $N = 4,5,6$ but not at $N = 3$:

$$\boxed{\Delta\beta_3^{\rm pen} = -\frac{25}{64} = -\frac{15}{16}\alpha_3^{\rm pen}, \qquad \Delta\alpha_3^{\rm pen} = 0.}$$

Equivalently,

$$\boxed{\Delta C_3^{\mathrm{shp}} = -\frac{25}{1024}}$$

with $A_3^{\mathrm{shp}}, B_3^{\mathrm{shp}}, D_3^{\mathrm{shp}}$ unchanged. The antisymmetric ladder truncates at $SU(3)$ because $\Lambda^3 V$ is already the singlet; the stable channel content requiring $\Lambda^4 V$ is absent.

Therefore the broad quotient-scalarity statement

$$P_{\mathrm{cage}} H^{\mathrm{det}} P_{\mathrm{cage}} = \delta q_N P_{\mathrm{cage}}$$

fails at $N = 3$. It is certified at $N = 4$ and $N = 6$, vacuous at $N = 5$, and any conjectural generalization must be restricted to $N \geq 4$.

1.6.9 4.9 Exact coefficient anchors and provenance

v1.4 proof-status note. The rational $SU(3)$ values below are retained exactly, because the independent audit reproduces the same band edges, bandwidth, and directional curvatures. They are now classified as an **exact-arithmetic computational ledger — audit pending**, not as a cold end-to-end computer-assisted theorem. The audit challenges certificate completeness, not the reproduced rational values.

For $SU(3)$,

$$q_3 = -\frac{20721577909065127111}{7250590288602460800}, \qquad \alpha_3^{\mathrm{pen}} = \frac{5}{12},$$
$$\beta_3^{\mathrm{pen}} = \frac{17607806155349}{275331901291200}.$$

Thus

$$(A_3^{\mathrm{shp}}, B_3^{\mathrm{shp}}, C_3^{\mathrm{shp}}, D_3^{\mathrm{shp}}) = \left(\frac{5}{48}, 0, -\frac{211835444920651}{4405310420659200}, 0\right).$$

For $SU(4)$,

$$q_4 = -\frac{162485785670299274695454289332603}{121294607143027203361265133093750},$$
$$\alpha_4^{\mathrm{pen}} = \frac{32}{675}, \qquad \beta_4^{\mathrm{pen}} = \frac{3601925923737103752887}{70481696720359496343750}.$$

The exceptional matrix is not a scalar multiple of I_3 on the full one-flux space; the certified statement is the all-zone branch identity

$$H_{4,4}^{\mathrm{exc}}(k)\psi(k) = \Delta q_4 \psi(k).$$

For $SU(5)$ and $SU(6)$, the 2026-08-08 homological-flat-band paper promotes the finite-rank certificates into the current Layer-II status authority: α_5^{pen} and α_6^{pen} are exact as above, the second pencil coefficient agrees with the stable expression at both ranks, $SU(5)$ has no determinant sector, and $SU(6)$ has the exact rest correction 6/343. This master document does not invent unreproduced giant rational forms that are not printed in the supplied source.

1.6.10 4.10 Physical scope of the one-plaquette band

These theorems are fixed-lattice, finite-order statements inside the one-excitation truncation. They do not identify the caged one-plaquette state with the physical 1^{+-} glueball. The matched Monte Carlo audit instead finds the raw one-plaquette ImTr operator carries spectral weight 0.0072 ± 0.0165, while the smeared basis couples at amplitude about 0.80. The physical state is extended; the one-plaquette result is an exact operator/geometry seed, and physical completion runs through smearing and multi-plaquette dressing.

1.6.11 4.11 $SU(3)$ fourth-order audit firewall

The independent `SIMULATIONS/` audit separates the $O(y^4)$ result into what is already algebraically secure and what still requires certificate closure.

1.6.11.1 Retained exact computational ledger

The audit independently reproduces

$$c_4(\Gamma) = -\frac{20721577909065127111}{7250590288602460800},$$

$$\boxed{\Delta c_4 = \frac{132329431693349}{275331901291200} \approx 0.48061786909826,}$$

and the directional curvatures

$$\kappa_{100} = \frac{5}{24}, \qquad \kappa_{110} = \frac{247051057231349}{2202655210329600}, \qquad \kappa_{111} = \frac{132329431693349}{1651991407747200}.$$

The exact shape coefficient $A = 5/48$ and the two consistency identities among the ray curvatures are also reproduced. These values remain the canonical computational ledger.

1.6.11.2 Six theorem-closure gates

1. **Model-space scalar gate:** verify rather than assume
$$PVP = aP.$$
2. **Degenerate folded-formula gate:** regression-test the folded des-Cloizeaux effective Hamiltonian on a genuinely multidimensional degenerate model space, including off-diagonal hopping.
3. **Independent-regression gate:** extend Stage 3H from 1,478 to all 3,895 topologies, including the 2,417 mixed/all-resonant folded cases.
4. **Magnetic-normalization gate:** prove the convention-dependent plaquette prefactor already absorbed into the perturbative variable and lock a source-to-canonical normalization identity.
5. **Interval-rigor gate:** remove the point-to-interval wraps for `theta`, `delta`, `plo/phi`, and tile the Brillouin zone using outward-rounded π.
6. **Touching/uniformity gate:** control the region $|k| \lesssim y$ where the $O(y^2)$ branch separation vanishes as $y^2|k|^2$ and becomes comparable to the $O(y^4)$ correction.

Two additional reproducibility tasks are mandatory before archival theorem status: ship `y4_full_real_space_H4_kernel.js` with a reference SHA-256, and correct the downstream use of the rigid -2.5134 component where the physical band minimum requires $c_4(\Gamma) \approx -2.857916$.

1.6.11.3 Status consequence

Until these gates close, the strongest defensible wording is

$$\boxed{SU(3) \text{ first dispersion at fourth order: exact-arithmetic computational result, heavily cross-checked.}}$$

It is **not yet** a cold, end-to-end computer-assisted theorem. In particular, the interval code's current `proved=true` flag should be read as decisive numerical certification of the implemented function, not as fully rigorous interval enclosure of the continuous Brillouin zone.

1.6.11.4 Extrapolation firewall

The short strong-coupling series is not controlled as a continuum extrapolation. The available five-term Borel-Pade estimates are consistent with the lattice scale but do not converge tightly enough to support a parameter-free continuum glueball prediction. The correct status remains

$$\boxed{\text{consistent, not controlled.}}$$

1.7 5. Layer III — Seam analyticity (the strong/weak crossover)

1.7.1 5.1 The exceptional-point atlas — Computationally verified, two points Kantorovich-certified to 12 digits

For $H(\beta) = \mathrm{diag}(\frac{1}{2}C_2) - \frac{\beta}{6}(X + X^\top)$ on $SU(3)$ class functions (K-stable to 10^{-14} across $K = 16 \ldots 28$):

sector	β_c	$\lvert\beta_c\rvert$	angle	colliding pair
even	$+0.797842828512 + 1.389351779364i$	**1.6021**	60.13°	$E_0 \leftrightarrow E_1$ **(vacuum–gap EP)**
even	$-2.274880566451 + 0.838479039787i$	**2.4245**	159.8°	$E_1 \leftrightarrow E_2$
even	$-0.0036 + 0.3638i$; $-0.4737 + 0.7948i$; $+2.6993 + 0.2303i$	—	—	E_2E_3; E_4E_5; E_3E_4
odd	$+0.4578 + 0.8204i$; $-0.0332 + 1.6904i$; $+2.2554 + 2.1895i$	—	—	O_2O_3; O_4O_5; O_0O_1

The two governing EPs are certified via the bordered complex-symmetric system $(H - E)v = 0$, $\ell^\top v = 1$, $v^\top v = 0$ (residuals $\sim 10^{-16}$, Kantorovich $h < 1/2$ with large margin; interval arithmetic and truncation-tail bounds queued for full rigor).

1.7.2 5.2 The odd-sector structure theorem — Proven mechanism + machine-verified

Because the dominant singularity is a vacuum–gap collision, symmetric functions of the pair are analytic there. With $S = E_0 + E_1$, $G := O_0 - S/2$:

$$\boxed{\Delta^-(\beta) = G(\beta) + \tfrac{1}{2}\Delta^+(\beta), \qquad G \text{ analytic on } |\beta| < 2.4245.}$$

Direct verification at β_c: G's branch residual 3.0×10^{-5} vs raw Δ^-'s 1.5×10^{-1}. Consequences: the odd tower's radius is **inherited entirely from even-sector physics** (the vacuum EP at 1.6021, not O_0's own first EP at 3.14); squaring removes the E_0E_1 branch points, so $(\Delta^+)^2$ converges on $|\beta| < 2.4245$ — the exact analytic reason the manuscript's Theorem-4.2 closure $\Delta^+ = \sqrt{\frac{1}{2} + \frac{2}{3}\beta\, g(\beta)}$ reaches 9.5×10^{-4} while every raw Padé stalls at 6.1×10^{-2}; 2.4245 is its hard ceiling (obstruction: the E_1E_2 collision at negative coupling). Three independent consistency checks (level-repulsion minimum near $\beta \approx 0.8$; period-6 sign pattern of $b_1 \ldots b_6$ predicting 60°; root-test/seam profile) all agree.

1.7.3 5.3 Two-sector certified crossover v1 — Computationally verified

With exact strong towers through β^6 and the exact weak identity $G \sim \sqrt{2\beta/3} - \frac{7}{32} - \frac{1271\sqrt{6}}{55296}\beta^{-1/2} - \frac{7903}{221184}\beta^{-1}$ (leading term exact: $\sqrt{3\beta/2} - \frac{1}{2}\sqrt{2\beta/3} \equiv \sqrt{2\beta/3}$), pole-free two-point rationals in $x = \sqrt{\beta}$ on $\beta \in [0.25, 50]$ achieve: $(\Delta^+)^2 \to \Delta^+$: 3.3×10^{-2}; G: 1.3×10^{-2}; assembled Δ^-: 1.9×10^{-2} — the odd sector's **first controlled crossover**. Recorded failure mode: the maximal exactly-determined system amplified a 10^{-3} tower inconsistency into 84% error; standing rule — **gate the fit residual, not just pole-freeness**.

1.8 6. Layer IV — Transfer geometry and certificates

1.8.1 6.1 Wilson–Bergman weight theorem — Proven (+8/8 gates)

The weight $w_\lambda(\beta) = \int_{SU(N)} |\chi_\lambda|^2 e^{\beta((2/N)\,\mathrm{Re}\,\mathrm{tr}\, g - 2)} dg$ is a Bessel–Toeplitz determinant:

$$w_\lambda(\beta) = c_N(\beta) \sum_{s \in \mathbb{Z}} \det\left[I_{\ell_a - \ell_b + s}(\kappa)\right]_{a,b=1}^{N}, \qquad \ell = \lambda + \rho, \quad \kappa = \frac{2\beta}{N}.$$

Theorem A (repaired): $\|M_q\| = \operatorname{ess\,sup} q = q(1) = (2N)^k$ despite the Weyl density vanishing to order $2|\Phi^+|$ at the maximizer — and the supremum is **not attained**: shell-K truncations obey $(2N)^k - \lambda_{\max}(K) \asymp C/K^2$ (Laplace mechanism; measured exponent 1.91 over $K = 48 \to 96$). Practical license: shell-K transfer estimates may use $\lambda_{\max}(K)$ (e.g. 1208 vs 1296 at $K = 24$). The earlier "N-uniform $\|M_q/q(1)\| = 1$" claim is **Withdrawn** as vacuous.

1.8.2 6.2 Bergman-transfer closure at ranks one and two — Computationally verified

The ledger's radial-tail no-go prescribed "a different transfer norm." Its concrete instantiation — compact-alcove character basis + Wilson–Bergman weights, matrix entries as Bergman-norm ratios of adjacent weight-lattice points — **closes the tail** with no n^2 growth: at rank one (SU(2) toy) and at rank two (SU(3) chamber), after quasi-degenerate blocking of the C-conjugation strips with fixed threshold $|\Delta C_2| \leq 1$ (provably terminating; max 5 band partners; min nonresonant gap exactly 4/3); $\sup T = 457, 593, 631$ at $\beta = 4, 16, 64$. The quartic Wilson object has exact integer character entries, exact bandwidth 4, exact deep-chamber translation invariance (rank-two linearization constant 90). The resonant δ-blocks are exactly where a real PMBSF secular analysis must live.

1.8.3 6.3 Exact computer-assisted certificate stack — Proven — exact computer-assisted

$SU(3)$ fourth-order real-space SOS theorem (positive local sum of squares); Stage-2A local projector theorem; Stage-3G targeted reduction; full-symbolic $N \geq 7$ walled-Brauer certificate with independent audit (all exact gates pass; runtime provenance limitation only); 387 lower-order hard gates rerun cold; hashed 189-record kernel and payload identities; the 2026-08-01 21-gate seam suite. Certificates are identified by SHA-256 and never overridden by prose.

1.8.4 6.4 Limitation theory (Direction 3) — Program, pre-registered

Template: transplant fixed-degree sign-uncertainty limitation theorems (Cohn–Dong–Gonçalves-type) through the compact-to-Euclidean contraction of the two-cone certificate geometry, with effective dimension $\lambda_{\text{eff}} \sim (N^2-1)/2$ and the quartic Weyl barcode $(N^2-1)(N^2-4)(N^2-9)/[4(N^2+1)]$ counting the opening angular channels. Kill criterion: if the Mellin multiplier of the $u^{\alpha_N} e^{-u}$ weight is not of Γ-ratio type, the transplant fails structurally. Established substrates for the positive side: Ozawa / Netzer–Thom / Kaluba–Nowak–Ozawa group-ring SOS; DFPR Lie–Schwinger block-diagonalization for the KS gap (its finite-local-dimension hypothesis is the single obstruction to re-prove for $L^2(SU(3))$ links).

1.9 7. Layer V — Marginal stability and projected capacity

1.9.1 7.1 Curvature sources — Proven

Root second-moment identity $\sum_\alpha \alpha \otimes \alpha = 2h^\vee I$; Weyl potential curvature floor $\nabla^2[-\log|\Delta|^2] \succeq h^\vee I/2$, exactly $N/2$ on $\mathbf{1}^\perp$ for $SU(N)$; exponential-coordinate Haar Hessian $h^\vee I/6$ at the identity. Fixed-seed Hessian checks exceed the floor for $N = 3, 4, 5, 8, 10$ (**CV, reproduced**).

1.9.2 7.2 Exact marginalization — Proven

Helffer–Sjöstrand covariance representation on the conditional fiber; leakage bound $\nabla_x^2 S_{\text{eff}} \succeq \mathbb{E}\nabla_{xx}^2 S - M^2 I/\gamma$; coercivity form $K_0 - V \succeq (1-\Theta)\rho_0 I$ with $\rho_0 = \sigma - M^2/\gamma > 0$, $\Theta < 1$.

1.9.3 7.3 Decay machinery

Finite-range Combes–Thomas inverse bound with conservative rate $\eta_{\text{CT}} = R^{-1}\log(1 + a_0/(2B_0))$ (**Proven**); for the 4d co-plaquette metric $a_0 = m^2$, $B_0 = 18\alpha$, $R = 1$, $C_0 = D_E = 18$. The sharper Davies–Gaffney $2\,\text{arsinh}(m/2\sqrt{\alpha k})$ rate remains **Conjectural** (finite $L = 16$, $d = 4$ certificate only, phase-corrected, max ratio 0.1411851688849).

1.9.4 7.4 Capacity: the correct global object — mixed status, sharply resolved

- Canonical incidence-shadow capacity $\mathcal{K}_{\Lambda,L}(\Gamma) = \|P\mathbf{1}_{C(\Gamma)}P\| \le 1$ and the rooted polymer bound with **one** mark $\mathfrak{Z}_{p_0} \le C_{\mathcal{P}} e^{s\gamma}\, \mu z e^a/(1-\mu z e^a)$, Peierls activity $z = K_\alpha e^{-(1-\alpha)\beta\delta}$ — **Proven as implications**.
- **Global fixed-window firewall is false**: for Bernoulli defects of fixed density the projected top norm tends to one (**Proven no-go**). The correlated Wilson analogue of the rare-island result is **Conjectural/Open**.
- Caged-band capacity sandwich $P^G_{\Lambda/c_+} \preceq P^M_\Lambda \preceq P^G_{\Lambda/c_-}$ — **Proven** from sharp mobility ellipticity: every positive projected support capacity of the caged band is squeezed between ordinary lattice-Laplacian capacities.
- PC chain: source-tilt identity **Proven** (PC-1); Peierls, square-free Wilson-to-Bernoulli domination, rooted summability **Proven implications** (PC-3,4,6); the two load-bearing hypotheses — **Wilson free-energy stability $Z_{\beta,\alpha,\Gamma,L}/Z_{\beta,L} \le K_\alpha^{|\Gamma|}$ (PC-2)** and the **source-radius reduction** — are **Open**.
- PMBSF conditional spine: $\boxed{\text{(ML)} \Rightarrow \text{Wilson-typical projected capacity} \Rightarrow \text{finite-lattice projected coercivity}}$; matrix-Laplace domination (ML) is **Open**; the earlier Wilson-to-block "proof" is **Withdrawn**; deterministic block lemmas survive; v7 fixed-window SU(2) comparisons are the highest-authority computational evidence; BM(q) vs EC(q,ℓ) are distinct properties requiring a defect-localization bridge (**Open** O5).

1.9.5 7.5 The PMBSF SU(2) conditional paper (v1.1) — the probability side as a theorem stack

The vague instruction "prove Lemma Q" is replaced by a proven reduction chain with two named analytic inputs. Guiding deterministic objects: projected plaquette atoms $A_p = P\mathbf{1}_{\partial p}P$ and the Birman–Schwinger criterion $\Theta_D = \|M^{-1/2}PV_DPM^{-1/2}\| < 1$ (finite-dimensional, unconditional). Central probabilistic input, **Lemma Q** (conditional rare-source factorization, with rooted version):

$$\mathbb{E}\left[\prod_{p\in B} X_{p,\eta} \mid \mathcal{F}_{C^c}\right] \le (C_Q q_\eta)^{|B|}.$$

Proven interface:

$$\mathrm{LCI}_{\mathrm{good}} + \mathrm{BFS}_{\mathrm{far}} \Rightarrow \mathrm{TOS} + \mathrm{J} \Rightarrow Z_A(\rho/q_\eta) \le e^{K|A|} \Rightarrow \text{Lemma Q}.$$

$$\text{Lemma Q} + \mathrm{SWB} + \mathrm{BBG} \Rightarrow \text{firewall closure}.$$

The **positive source-radius pivot** replaces fragile product-moment/mixing arguments and needs no complex zero-free region: $Z_A(s) = \sum_{R\subset A} s^{|R|}\mathbb{E}\prod_R X_p$ is a factorial-moment generating function with nonnegative coefficients, so $s^{|B|}\mathbb{E}\prod_B X_p \le Z_B(s)$ (strict when $q_\eta > 0$); the exact per-step ratio $Z_{A\cup\{p\}}/Z_A = 1 + s\,\mathbb{E}_{\mu^{A,s}} X_p$ telescopes under an exponentially summable influence kernel ($J_* < \infty$) to $K = \rho\, C_{\mathrm{TOS}} e^{J_*}$, with canonical optimum $\rho_{\mathrm{opt}} = 1/(C_{\mathrm{TOS}}e^{J_*})$, $C_Q^{\mathrm{opt}} = e\, C_{\mathrm{TOS}} e^{J_*}$ (convex in ρ). Rooted bounds are soft conditionings costing one e^{J_*} per root. **Local geometry is exact:** the SU(2) one-link conditional law is $\mathrm{vMF}_4(\overline{H}_e/\|H_e\|, \beta\|H_e\|)$ on S^3; plaquette scores are linear, $\frac{1}{2}\operatorname{Re}\operatorname{Tr} U_r = u\cdot n_r$, so incident sources are spherical-cap intersections — $\mathrm{LCI}_{\mathrm{good}}$ is a finite-dimensional S^3 cap-intersection theorem for a log-concave density. The far field enters only through $\mathrm{BFS}_{\mathrm{far}}$ (Bałaban/Dimock source-weighted far stability with $J(p,r) \le C_J e^{-m_J d_C(p,r)}$). **Open inputs of this stack:** $\boxed{\mathrm{LCI}_{\mathrm{good}}}$ and $\boxed{\mathrm{BFS}_{\mathrm{far}}}$ (plus the separate SWB, BBG gates). Diagnostics at the working point $\beta = 3.5$, $q_\eta = 0.003$, $\eta = 0.005$ (frozen-block heat-bath through side-10, full-volume covariance through $L = 64$) are motivational only. The compact SU(3) remainder proof explicitly does **not** revive the archived global top-norm program; the two tracks stay firewalled.

1.10 8. Contact with physics: verification layer

- **Replay gate — verified.** The hardcoded Athenodorou–Teper T_1^{+-} table is a digit-for-digit faithful transcription of arXiv:2007.06422 (all 25 verifiable cells match); continuum $M(1^{+-})/\sqrt{\sigma} = 6.065(40) \approx 2.944(42)$ GeV, cross-consistent with Morningstar–Peardon and Chen et al.
- **Like-for-like Monte Carlo — passed (23/23 hard gates, 21-gate suite green).** Hardened spatial MC at matched volume and coupling: $aM(T_1^{+-}) = 1.6897 \pm 0.121$ vs published 1.591(18) — pull +0.82; string scale $+0.71\sigma$; conditioning fix at $7.97 \times 10^3 < 10^4$ spec; throughput 7.1×10^5 site-sweeps/s. **The scientifically loaded number:** the raw single-plaquette ImTr operator carries 0.0072 ± 0.0165 ($< 4\%$ at 2σ) of the physical glueball, while the smeared basis couples at amplitude 0.80 — the physical T_1^{+-} state is extended. This converts the program's scope firewall (one-plaquette bridge = operator identity, physical completion runs through smearing) from precaution into **measured fact**.
- **Peter–Weyl and coordinate audits.** Compact $SU(3)$ audit confirms the three-term law (AUD-1); exact rational Wick/Gram closures pass for $N = 3 \dots 12$ even and odd (AUD-2/3); corrected GPU Weyl-triangle solver exists but **no CUDA execution certificate** yet (AUD-4 Open); the 21-gate suite validates triangle/character agreement (the Weyl-denominator conjugation numerically).
- **Structure predictions.** The shell decomposition (§4.6) reproduces the lattice's channel assignments at shell 4 and predicts the exotic C-odd ordering questions (e.g. 0^{--} vs 3^{+-}) as controlled shell-6 splittings — structure, not masses.

1.11 9. Master dependency graph

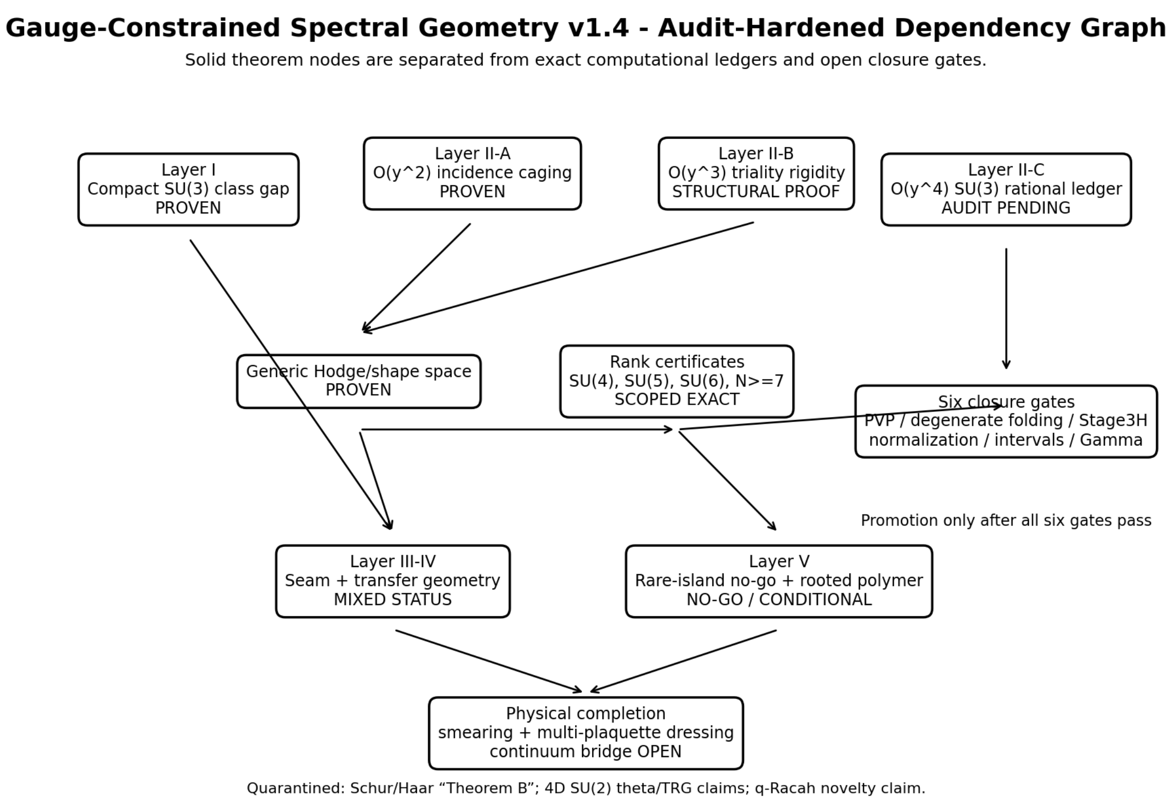

Figure 1: Master dependency graph for the five-layer theory

The dependency graph separates proved structural nodes, scoped exact certificates, audit-pending computational ledgers, and open closure gates; arrows denote interfaces and dependencies, not automatic theorem promotion.

1.12 10. Canonical constants index (selected)

1.12.1 10.1 Couplings and local class geometry

- **Canonical strong-coupling variable:** $u = \beta_H/6$. **Status:** Proven bridge.
- **Weyl-Gaussian radial parameter:** $\alpha_N = (N^2-3)/2$. **Status:** Proven.
- **Compact $SU(3)$ even gap:** $c_0^+ = -5/16$, $c_1^+ = -311\sqrt{6}/9216$. **Status:** Proven.
- **First non-radial escape:** $\sqrt{6}/576$. **Status:** Proven.
- **Fixed-rank constants:** $c_0^{(N)} = -(2N^2-3)/(16N)$ and $c_0^{(N),-} = -3(N^2-3)/(16N)$. **Status:** Proven.
- **Polarity excess:** $9/(32N)$. **Status:** Proven corollary.
- **Exact $SU(3)$ next coefficients:** $c_2^+ = -5665/110592$, $c_1^- = -551\sqrt{6}/13824$, $c_2^- = -53/864$. **Status:** Exact arithmetic; the higher-order analytic remainder remains separate.

1.12.2 10.2 Strong-coupling mobility

- **Second-order rank scalar:**

$$t_N = \frac{2N(N^2-4)}{(N^2-1)(2N^2-1)(4N^2-9)}, \qquad N \geq 3.$$

  Status: Proven.
- **Flat-space dimension on T_L^3:** L^3+2. **Status:** Proven.
- **First level above the flat carrier:** $4\sin^2(\pi/L)$. **Status:** Proven.
- **Universal fourth-order axial pencil coefficient:**

$$\boxed{\alpha_N^{\mathrm{pen}} = \frac{640}{N(N^2-1)^3}}, \qquad N \geq 3.$$

  Status: Exact certificate-backed rank identity; end-to-end theorem promotion inherits any shared folded-effective-Hamiltonian/normalization gates identified by the v1.4 audit.
- **Exceptional axial values:** $\alpha_3 = 5/12$, $\alpha_4 = 32/675$, $\alpha_5 = 1/108$, $\alpha_6 = 64/25725$.
- $SU(3)$ **second-pencil anomaly:** $\Delta\beta_3^{\mathrm{pen}} = -25/64$, equivalently $\Delta C_3^{\mathrm{shp}} = -25/1024$. **Status:** Exact relative to the accepted finite-rank/stable certificates.
- $SU(4)$ **exceptional rest shift:** $\Delta q_4 = -304746539168/160249753125$. **Status:** Proven relative to the narrow $SU(4)$ exceptional certificate; shared upstream folded-formalism dependence should be stated if applicable.
- $SU(5)$ **determinant sector:** empty. **Status:** Exact finite-rank scan in the 2026-08-08 Layer-II authority.
- $SU(6)$ **exceptional rest shift:** $\Delta q_6 = 6/343$. **Status:** Exact finite-rank certificate as summarized in the 2026-08-08 Layer-II authority.
- **Physical tier collapse:** $B_N^{\mathrm{shp}} = D_N^{\mathrm{shp}} = 0$ for every resolved rank, now covering the complete rank partition $N \geq 3$. **Status:** Exact within accepted coefficient ledgers; theorem-grade promotion is audit-sensitive where a rank shares the unclosed fourth-order effective-Hamiltonian pipeline.
- $SU(3)$ **pencil:** q_3 as in §4.9, $\alpha_3^{\mathrm{pen}} = 5/12$, $\beta_3^{\mathrm{pen}} = 17607806155349/275331901291200$. **Status:** Exact-arithmetic computational ledger — audit pending; values independently reproduced, six theorem-closure gates remain.
- $SU(4)$ **pencil:** q_4 as in §4.9, $\alpha_4^{\mathrm{pen}} = 32/675$, $\beta_4^{\mathrm{pen}} = 3601925923737103752887/7048169672035949634375$. **Status:** Proven - exact computer-assisted.

1.12.3 10.3 Seam, transfer, and global diagnostics

- **Vacuum exceptional point:** $0.797842828512 + 1.389351779364i$. **Status:** Kantorovich-certified at the recorded truncation.

- **Squared-gap analytic radius:** 2.4245, controlled by the $E_1 \leftrightarrow E_2$ exceptional point. **Status:** Computationally verified.
- **Wilson-Bergman multiplier norm:** $\|M_q\| = (2N)^k$, with truncation deficit $\asymp C/K^2$. **Status:** Proven plus verified.
- **Finite-channel leakage Perron root:** $0.5501615335\ldots$. **Status:** Computationally verified; not a full-channel constant.
- **Combes-Thomas data in four dimensions:** $a_0 = m^2$, $B_0 = 18\alpha$, $C_0 = 18$. **Status:** Proven.
- **Matched Monte Carlo mass:** 1.6897±0.121 versus 1.591(18), pull +0.82. **Status:** Computationally verified.
- **Raw single-plaquette overlap:** 0.0072 ± 0.0165. **Status:** Measured.

1.13 11. The firewall: what the master theory does not claim

No result in this theory establishes, and none is presented as establishing: convergence of the strong-coupling series beyond $|\beta_c|$; a continuum 1^{+-} glueball mass or a controlled $m_{1^{+-}}/\sqrt{\sigma}$; a pseudoreal $SU(2)$ extension of the $N \geq 3$ mobility theorem; an all-orders localization theorem; infinite-volume Wilson exponential clustering; identification of local class gaps with glueball masses; confinement or a string tension from the present coefficients; Osterwalder–Schrader reconstruction; a valid 4d $SU(2)$ θ-vacuum partition function or topological susceptibility; or the four-dimensional Yang–Mills mass gap. The strong-coupling mobility theorems, the compact large-β class asymptotics, and the global Wilson-measure program are three separately controlled regimes; the theory's unification is of **structure and method**, and each proved formula is confined to its regime.

1.14 12. Open frontier: the load-bearing inputs and pre-registered tests

Layer-II frontier after the 2026-08-08 closure: the axial fourth-order coefficient is no longer an open-rank problem. The immediate mobility questions are (i) whether the b_2-fold protected level survives coupling to multi-plaquette sectors, (ii) a clean cold rerun of the named third-order $SU(3)$ certificate chain before submission, and (iii) extension of the Betti-count theorem to noncubic cellulations with independently tunable b_2. The $SU(5)/SU(6)$ axial closure and the $SU(3)$ second-pencil anomaly are now part of the theorem registry, not frontier items.

Seven inputs on which the next tier of the theory turns: 1. **Symbolic-N Gram transcript** for the even/odd inverse-Gram resolvent identities -> promotes the $c_1^{(N),\pm}$ **and the new** $c_2^{(N),\pm}$ candidates to unrestricted fixed-rank theorems. 2. **Compact Casimir-shell buffered Schur-complement theorem** (SU3-10) -> the surviving route to a genuine full-channel local constant. 3. $\mathrm{LCI}_{\mathrm{good}}$ + $\mathrm{BFS}_{\mathrm{far}}$ — the PMBSF SU(2) paper's two named analytic inputs (they imply TOS+J -> positive source radius -> Lemma Q), together with the separate SWB and BBG gates; on the Peierls route, **PC-2 Wilson free-energy stability** with useful $\log K_\alpha$ plus the **source-radius reduction** remain the parallel open pair. 4. **Matrix-Laplace domination (ML)** or a replacement giving the same Wilson projected-capacity tail -> discharges the PMBSF conditional spine. 5. **Certified crossover v2** (extended exact towers via validated Cauchy extraction; conformal-map two-point builds with residual gating; interval-rigorous EP certificates including truncation tails) -> Theorem-4.2-class $10^{-3}-10^{-4}$ accuracy in **both** sectors — now feedable with the four-term weak towers of §3.2b. 6. **The KS finite-dimensionality re-proof** in the DFPR Lie–Schwinger scheme for $L^2(SU(3))$ links -> an explicit-threshold infinite-volume KS gap theorem, which would be new to the literature. 7. **One more order of the equivariant harmonic-well lemma** (quasimode residual $O(h^5)$) -> promotes the exact c_2 values to a proven $O(\beta^{-3/2})$ remainder; and **closing the small-rank verification gaps** (even $N = 4, 5$; odd $N = 4, 5, 6$) via rank-reduced invariant bases, as done for $SU(3)$ with $\{p_2, p_3\}$.

Pre-registered tests with kill criteria (2026-08-01 ordering): - T1(a): derive the Mellin multiplier of the $u^{\alpha_N} e^{-u}$ weight; if not of Γ-ratio type, the limitation transplant downgrades to method-inspiration.

Then run the Direction-3 primal at $N = 3, 4, 6$ against the $(1/\pi)\sqrt{d}$-type localization scaling. - **T3 + audit item 3:** re-run the A' budget with amortized-KL/coupling accounting and the sharpened incidence constant 16; require $\geq 100\times$ improvement or drop the direction. - **T2:** exponential-tilt check between exact strong and weak towers (compact-optimizer -> Euclidean-optimizer degeneration), feeding crossover v2. - **T4 continuation:** SU(N) chamber generalization of the Bergman blocking (conjugation-strip $(N-1)$-torus; block sizes conjectured N-independent — checkable), then recast the actual Lemma-Q/A' tails in the Bergman norm with $\delta = 1$ bookkeeping. - **MC campaign, staged:** three next-coarsest ensembles (~3 A100-days) -> four-point continuum fit; re-anchor at 26^3; overlap-vs-APE-level flow measurement of the raw-fraction number.

1.15 13. Closing statement

The theory's core is a matched pair of exact escape theorems — $\sqrt{6}/576$ internally and the exact fourth-order mobility pencil spatially — together with the physical tier collapse $B_N^{\mathrm{shp}} = D_N^{\mathrm{shp}} = 0$ across the complete resolved rank partition $N \geq 3$ and the universal axial law $\alpha_N^{\mathrm{pen}} = 640/[N(N^2-1)^3]$ — with the internal tower now certified three corrections deep in both C-sectors and its analytic remainder proved in-corpus, welded by the Hodge complex's kernel–resolvent duality, extended along the coupling axis by an EP-certified analytic seam, equipped with a working transfer norm (Wilson–Bergman) and an exact certificate stack, and pointed at the global regime through a rooted capacity calculus whose two open hypotheses are precisely named. Everything Proven here is regime-scoped and hash-anchored; everything Open is listed with the experiment that would close or kill it. That discipline — filtration, rigidity, exact first escape, and an honest firewall — **is** the master theory.

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2 Appendices: detailed proof and certificate insertions

2.1 Appendix 0. Audit quarantine: results excluded from the master theorem graph

The following threads are explicitly **not** premises of Gauge-Constrained Spectral Geometry v1.4.

1. **Schur/Haar-Hessian "Theorem B".** The claimed bound $\rho_{\mathrm{eff}} \geq N/2 - M^2/\gamma$ is rejected in its published-notebook form: its Schur-complement positivity precondition fails on a substantial set, the Haar Hessian is unbounded below near eigenangle collisions, median-based clipping cannot substitute for a uniform lower bound, and the reported N-growth is imposed by the chosen $\beta \propto N$ scan. The separately derived analytical Wilson+Haar Hessian formula is salvageable as a technical calculation, but it does not prove a lattice spectral gap.
2. **4D $SU(2)$ θ/TRG thread.** The audited implementation does not yet encode the Wilson plaquette action, does not implement the claimed θ dependence, and does not perform a validated 4D coarse-graining. It is excluded from all physics claims. The underlying Levin-Nave `trg_step` kernel is salvageable when supplied with a correct benchmark tensor.
3. **q-Racah/Doob gap novelty claim.** The clean implementation is correct but reproduces the standard $n = 1$ q-Racah/Askey-Wilson eigenvalue and therefore is treated as a validation/utility result, not a new theorem.

2.2 Appendix A. Compact $SU(3)$ class-gap remainder closure

The following proof is the detailed analytic insertion underlying §3.1. It is reproduced from `SU3_Compact_Remainder_Proof.docx`. The exact reduced-resolvent value

$$\Delta_{\mathrm{res}} = -\frac{205\sqrt{6}}{3072}$$

is inherited from the exact oscillator coefficient ledger; the accompanying verification script checks the invariant identities, moments, direct sextic term, and final assembly, but does not independently reconstruct that reduced-resolvent matrix element.

Status: rigorous proof insertion for the local theorem
Scope: fixed-rank $SU(3)$, charge-conjugation-even class sector, $\beta \to \infty$
Not claimed: an infinite-volume lattice gap, a glueball mass, or a Yang-Mills mass gap

2.2.1 1. Result

Let

$$H_\beta = \frac{1}{2}C_2 + \beta\left(1 - \frac{1}{3}Re\chi_{(1,0)}(g)\right)$$

act on square-integrable class functions on $SU(3)$, with

$$C_2(p,q) = \frac{1}{3}\left(p^2 + pq + q^2\right) + p + q.$$

Let $E_0^+(\beta)$ and $E_1^+(\beta)$ denote the two lowest eigenvalues in the charge-conjugation-even class sector, ordered by

$$E_0^+(\beta) < E_1^+(\beta).$$

Define

$$\Delta_{SU(3)}(\beta) = E_1^+(\beta) - E_0^+(\beta).$$

Then there are constants $C < \infty$ and $\beta_0 < \infty$ such that, for every $\beta \geq \beta_0$,

$$\boxed{\left|\Delta_{SU(3)}(\beta) - \left(\sqrt{\frac{2\beta}{3}} - \frac{5}{16} - \frac{311\sqrt{6}}{9216}\,\beta^{-1/2}\right)\right| \leq C\beta^{-1}.}$$

This closes the analytic remainder missing from the coefficient calculation.

2.2.2 2. Exact reduction to a flat torus Schrödinger operator

Let $\mathfrak{t} \cong \mathbb{R}^2$ be the trace-zero Cartan plane with orthonormal coordinates (x, y),

$$\theta_1 = \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{6}}, \qquad \theta_2 = -\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{6}}, \qquad \theta_3 = -\frac{2y}{\sqrt{6}}.$$

Let $T = \mathfrak{t}/\left(2\pi Q^\vee\right)$ be the maximal torus and let

$$J(\theta) = \prod_{\alpha>0} 2\sin\frac{\alpha(\theta)}{2}$$

be the real Weyl denominator. Weyl integration shows that multiplication by J is unitary, up to an irrelevant normalization, from class functions to Weyl-anti-invariant functions in flat $L^2(T, d\theta)$.

For an irreducible character χ_λ, the Weyl character formula gives

$$J\chi_\lambda = A_{\lambda+\rho},$$

where $A_{\lambda+\rho}$ is the alternating torus exponential. Since

$$-\Delta_{\mathfrak{t}} A_{\lambda+\rho} = |\lambda+\rho|^2 A_{\lambda+\rho}$$

and

$$C_2(\lambda) = \frac{|\lambda+\rho|^2 - |\rho|^2}{2},$$

the character basis proves the exact operator identity

$$J\left(\frac{1}{2}C_2\right)J^{-1} = \frac{1}{4}\left(-\Delta_{\mathfrak{t}} - |\rho|^2\right).$$

For $SU(3)$, $|\rho|^2 = 2$. Consequently H_β is unitarily equivalent, in the appropriate Weyl-anti-invariant symmetry sector, to

$$\widehat{H}_\beta = -\frac{1}{4}\Delta_{\mathfrak{t}} - \frac{1}{2} + \beta W(\theta),$$

where

$$W(\theta) = 1 - \frac{1}{3}\sum_{j=1}^{3}\cos\theta_j.$$

This reduction is exact. In particular, there are no untracked metric or Haar-measure terms in the local expansion. The scalar $-1/2$ cancels from the gap.

2.2.3 3. The Wilson potential has one nondegenerate well

Because each $\cos\theta_j \le 1$,

$$W(\theta) \ge 0.$$

Equality requires $\cos\theta_j = 1$ for all j, hence $\theta = 0$ on T. Thus the identity is the unique global minimum. Near it,

$$W(\theta) = \frac{1}{6}P_2 - \frac{1}{72}P_4 + \frac{1}{2160}P_6 + O\left(|\theta|^8\right), \qquad P_k = \sum_{j=1}^{3}\theta_j^k.$$

Since $P_2 = x^2 + y^2$,

$$D^2W(0) = \frac{1}{3}I_2 > 0.$$

It follows that there are $r, c_0, c_1 > 0$ such that

$$W(\theta) \ge c_0|\theta|^2 \quad (|\theta| < r), \qquad W(\theta) \ge c_1 \quad (|\theta| \ge r).$$

These are precisely the confinement hypotheses needed for compact harmonic-well asymptotics.

2.2.4 4. Correct semiclassical parameter and scaled normal form

Set

$$h = \beta^{-1/2}, \qquad P_h = \beta^{-1}\widehat{H}_\beta = h^2\widehat{H}_\beta.$$

Then

$$P_h = -\frac{h^2}{4}\Delta_{\mathfrak{t}} - \frac{h^2}{4}|\rho|^2 + W(\theta).$$

This is a conventional semiclassical Schrödinger operator. Its low eigenvalues are $O(h)$. Use the well scaling

$$\theta = a\sqrt{h}\, z, \qquad a = \left(\frac{3}{2}\right)^{1/4}.$$

In the localized well, division by h gives

$$h^{-1}P_h = H_0 + hH_1 + h^2H_2 + h^3R_h,$$

where

$$H_0 = \frac{1}{2\sqrt{6}}\left(-\Delta_z + p_2(z)\right),$$

$$H_1 = -\frac{|\rho|^2}{4} - \frac{p_2^2}{96},$$

and

$$H_2 = \sqrt{6}\left(\frac{p_2^3}{11520} + \frac{p_3^2}{8640}\right).$$

The scale factors are exact:

$$a^4 = \frac{3}{2}, \qquad a^6 = \left(\frac{3}{2}\right)^{3/2} = \frac{3\sqrt{6}}{4}.$$

They give

$$-\frac{a^4}{144} = -\frac{1}{96}, \qquad \frac{a^6}{8640} = \frac{\sqrt{6}}{11520}, \qquad \frac{a^6}{6480} = \frac{\sqrt{6}}{8640}.$$

Here the $SU(3)$ Newton identities

$$P_4 = \frac{1}{2}p_2^2, \qquad P_6 = \frac{1}{4}p_2^3 + \frac{1}{3}p_3^2$$

have been used. On every fixed finite oscillator spectral subspace, R_h is uniformly bounded as $h \downarrow 0$. More explicitly, Taylor's theorem gives a multiplication remainder bounded by $C\langle z\rangle^8$, and every vector used below is a polynomial times a Gaussian.

The scalar term $-|\rho|^2/4 = -1/2$ in H_1 changes both energies equally, has no off-diagonal matrix elements, and therefore makes no contribution to the gap. Removing it recovers exactly the H_1 used in the coefficient tables.

2.2.5 5. Equivariant harmonic-well remainder lemma

2.2.5.1 Lemma Let P_h be the operator above, restricted to the Weyl-anti-invariant symmetry sector corresponding under multiplication by J to charge-conjugation-even class functions. Let $\mu_a^{(0)}$ be a simple eigenvalue of H_0 in that sector, with normalized eigenvector ϕ_a. Then the corresponding compact eigenvalue $p_a(h)$ of P_h satisfies

$$p_a(h) = h\mu_a^{(0)} + h^2\mu_a^{(1)} + h^3\mu_a^{(2)} + O\left(h^4\right),$$

where

$$\mu_a^{(1)} = \langle \phi_a, H_1\phi_a\rangle,$$

and

$$\mu_a^{(2)} = \langle \phi_a, H_2\phi_a\rangle + \sum_{b\neq a} \frac{\left|\langle \phi_b, H_1\phi_a\rangle\right|^2}{\mu_a^{(0)} - \mu_b^{(0)}}.$$

2.2.5.2 Proof Choose a Weyl-invariant cutoff χ supported in a fixed coordinate ball about 0 and equal to one on a smaller ball. The well bounds in Section 3 and the IMS formula imply that every eigenfunction of P_h with eigenvalue at most Ch has L^2-mass outside that smaller ball smaller than $O\left(h^N\right)$ for every fixed N; the usual Agmon estimate gives the stronger exponential bound. Thus the low spectrum is determined, to $O\left(h^N\right)$, by the Taylor germ at the unique well.

Let $Q_a = I - |\phi_a\rangle\langle\phi_a|$. Because $\mu_a^{(0)}$ is simple in the chosen symmetry sector, the reduced inverse

$$Q_a\left(\mu_a^{(0)} - H_0\right)^{-1} Q_a$$

is bounded on the finite polynomial-Gaussian sources generated at the orders used here. Define the first correction by

$$\phi_a^{(1)} = Q_a\left(\mu_a^{(0)} - H_0\right)^{-1} Q_a H_1\phi_a.$$

The order-h solvability condition gives

$$\mu_a^{(1)} = \langle \phi_a, H_1\phi_a\rangle.$$

Solving once more at order h^2 gives a polynomial-Gaussian vector $\phi_a^{(2)}$, and its solvability condition gives

$$\mu_a^{(2)} = \langle \phi_a, H_2\phi_a\rangle + \langle H_1\phi_a, Q_a\left(\mu_a^{(0)} - H_0\right)^{-1} Q_a H_1\phi_a\rangle.$$

Expanding the reduced inverse in the oscillator eigenbasis gives the displayed Rayleigh-Schrödinger sum.

After scaling back and applying χ,

$$u_{a,h} = \chi\, S_h^{-1}\left(\phi_a + h\phi_a^{(1)} + h^2\phi_a^{(2)}\right)$$

is a symmetry-preserving quasimode. The Taylor remainder contributes $O\left(h^3\right)$ to $h^{-1}P_h$, cutoff commutators are $O\left(h^N\right)$ for every N, and hence

$$\| \left[P_h - \left(h\mu_a^{(0)} + h^2\mu_a^{(1)} + h^3\mu_a^{(2)}\right)\right] u_{a,h} \| \leq Ch^4 \| u_{a,h} \| .$$

The harmonic approximation and the well bounds identify the low compact spectrum with the oscillator spectrum after division by h, including multiplicities. The relevant oscillator level is isolated and simple in the selected symmetry block, so the spectral theorem identifies one and only one compact eigenvalue in its $o(h)$ neighborhood. Its distance from the quasimode energy is $O\left(h^4\right)$, proving the expansion.

All operators, cutoffs, reduced inverses, and Riesz projections commute with the finite Weyl and charge-conjugation symmetries, so the standard harmonic-well construction restricts without change to the required sector. This is also a direct special case of the nondegenerate-well spectral equivalence and quantum Birkhoff normal-form results of Laurent Charles and San Vũ Ngọc, *Spectral asymptotics via the semiclassical Birkhoff normal form*, especially their comparison and normal-form theorems.

2.2.6 6. Application to the two $SU(3)$ class levels

Let

$$\delta(z) = \prod_{i<j} \left(\theta_i(z) - \theta_j(z)\right)$$

be the linear Weyl discriminant. Multiplication by δ identifies the invariant polynomial model with the Weyl-anti-invariant oscillator model. The flat oscillator states have the form

$$\phi_a(z) = \delta(z)\psi_a(z)e^{-p_2(z)/2},$$

and their flat inner product is exactly the project inner product

$$\langle f, g\rangle_W = \int_{\mathbb{R}^2} f(z)g(z)\delta(z)^2 e^{-p_2(z)}\, dz.$$

The first two charge-conjugation-even invariant factors are

$$\psi_0 = \text{constant}, \qquad \psi_1 \propto p_2 - 4.$$

They are the unique invariant states at relative shell degrees 0 and 2, so both model levels are simple in the required symmetry sector. Since

$$H_0 = \frac{1}{2\sqrt{6}}\left(-\Delta + p_2\right),$$

their leading energy difference is

$$\mu_1^{(0)} - \mu_0^{(0)} = \frac{2}{\sqrt{6}} = \sqrt{\frac{2}{3}}.$$

For the quartic term, the exact matrix elements give

$$\mu_1^{(1)} - \mu_0^{(1)} = -\frac{25}{48} + \frac{5}{24} = -\frac{5}{16}.$$

The direct sextic contribution is

$$\Delta_{H_2} = \frac{19\sqrt{6}}{576},$$

using

$$\langle p_2^3 \rangle_{\psi_0} = 120, \qquad \langle p_3^2 \rangle_{\psi_0} = 5,$$

$$\langle p_2^3 \rangle_{\psi_1} = 480, \qquad \langle p_3^2 \rangle_{\psi_1} = 20.$$

The exact reduced-resolvent contribution from H_1 is

$$\Delta_{res} = -\frac{205\sqrt{6}}{3072}.$$

Therefore

$$\mu_1^{(2)} - \mu_0^{(2)} = \frac{19\sqrt{6}}{576} - \frac{205\sqrt{6}}{3072} = -\frac{311\sqrt{6}}{9216}.$$

Applying the lemma to $a = 0, 1$ yields

$$p_1(h) - p_0(h) = h\sqrt{\frac{2}{3}} - h^2\frac{5}{16} - h^3\frac{311\sqrt{6}}{9216} + O\left(h^4\right).$$

Since

$$p_a(h) = h^2 E_a^+(\beta), \qquad h = \beta^{-1/2},$$

division by h^2 gives

$$\Delta_{SU(3)}(\beta) = \sqrt{\frac{2\beta}{3}} - \frac{5}{16} - \frac{311\sqrt{6}}{9216}\beta^{-1/2} + O\left(\beta^{-1}\right).$$

This proves the result in Section 1.

2.2.7 7. What has and has not been closed

The missing local analytic step is closed because:

1. Weyl conjugation is exact and leaves a flat kinetic operator plus a scalar.
2. The compact Wilson potential has a unique nondegenerate minimum.
3. The $h = \beta^{-1/2}$ rescaling produces the exact H_0, H_1, H_2 used by the coefficient calculation.
4. The two model levels are simple after the correct symmetry restriction.
5. A symmetry-preserving quasimode has residual $O\left(h^4\right)$ for P_h, which becomes $O\left(\beta^{-1}\right)$ for the original gap.

No explicit numerical value for C or β_0 is asserted. Obtaining effective constants would require a quantitative Agmon/IMS implementation, but is not necessary for the asymptotic theorem.

The proof does not repair or revive the projected-capacity program. That global program remains separate and archived under the audit's no-go conclusion for the top-norm observable.

2.3 Appendix B. Exact $SU(4)$ exceptional-rank completion

The following theorem is reproduced from `SU4_HYBRID_COMPLETE_THEOREM_V2 (2).md`. Its companion JSON certificate records the 4,171-word regression, 35,130 balanced fusion paths, 156 exceptional charge-conjugation orbits, exact local Gram reconstruction, 78 rank-one joint-channel factorizations, and the all-zone zero residual.

2.3.1 SU(4) fourth-order exceptional-rank completion

Status: PASS
Version: `2026-06-14-su4-hybrid-complete-v2`

2.3.1.1 Complete exceptional corpus The exact SU(4) N-ality scan introduces no new ordered words. The exceptional sector is contained in 76 of the existing 4,171 ordered words and consists of

- 312 exceptional sign assignments;
- 156 charge-conjugation orbits;
- 96 distinct exceptional trace topologies;
- 1,806 exact local-channel choices, of which 214 contract nontrivially.

The finite-rank local algebra contains 42 oriented exceptional signatures and 78 exact rank-one joint Casimir channels. The allowed final Haar families are

$$(4,0), \quad (0,4), \quad (5,1), \quad (1,5).$$

Determinant singlet channels occur in the resolvent only at the third des-Cloizeaux cut.

2.3.1.2 Exact correction on the flat branch The exceptional correction has 13 root-kernel entries and 39 cubic-completed real-space entries. It is **not** a scalar multiple of the identity on the complete three-component one-flux space.

Let

$$\psi(k) = \begin{pmatrix} e^{ik_2} - 1 \\ -(e^{ik_1} - 1) \\ e^{ik_0} - 1 \end{pmatrix}.$$

The exact Laurent-polynomial identity is

$$\boxed{H^{\mathrm{exc}}_{4,4}(k)\,\psi(k) = -\frac{304746539168}{160249753125}\,\psi(k)}$$

throughout the Brillouin zone. Hence the exceptional sector shifts the exact flat branch by the momentum-independent amount

$$\boxed{\Delta q_4 = -\frac{304746539168}{160249753125}}$$

while

$$\boxed{\Delta A_4 = \Delta B_4 = 0}.$$

2.3.1.3 Complete SU(4) coefficients

$$\boxed{q_4 = -\frac{16248578567029927469545428933260 3}{12129460714302720336126513309375 0}}$$

$$\boxed{A_4 = \frac{32}{675}}$$

$$\boxed{B_4 = \frac{3601925923737103752887}{70481696720359496343750}}$$

and therefore

$$\boxed{\Delta c_{4,4} = A_4 + B_4 = \frac{2314426811641505637629}{2349389890678649878125 0} > 0.}$$

The parity-point values obey

$$c_X = q_4 + A_4, \qquad c_M = q_4 + A_4 + \frac{1}{2}B_4, \qquad c_R = q_4 + A_4 + B_4,$$

and exactly

$$\boxed{c_R - 2c_M + c_X = 0}.$$

2.3.1.4 Full dispersion theorem For $k \neq \Gamma$, with $X_i = 1 - \cos k_i$,

$$\boxed{c_{4,4}(k) = q_4 + \frac{A_4 \sum_i X_i^2 + B_4 \sum_{i<j} X_i X_j}{2 \sum_i X_i}}$$

with continuous extension $c_{4,4}(\Gamma) = q_4$.

Because $A_4 > 0$ and $B_4 > 0$,

$$\boxed{\Gamma \text{ is the unique global minimum}}, \qquad \boxed{R \text{ is the unique global maximum}}.$$

Thus the SU(4) fourth-order one-flux T_1^{+-} band is strictly dispersive.

2.3.1.5 Verification chain

- canonical symbolic source SHA-256: `8feec874aa16c823bb837efa8df626d5cf735db5ecaa6c90b8806ddf456b51a5`;
- exact balanced rerun: 3,850 trace topologies and 35,130 fusion paths;
- stable 4,171-word corpus reproduced;
- complete exceptional scan: 76 words and 156 C-orbits;
- all explicit epsilon/delta-epsilon Gram matrices verified;
- all 78 joint channel projectors factorized and normalized exactly;
- exact all-zone Laurent-polynomial residual: `(0,0,0)`;
- corrected kernel support: 63 root entries and 189 real-space entries.

2.3.1.6 Correction to the preliminary interpretation The preliminary statement `Delta H_4,4(k)=Delta q_4 I_3` was too strong. The certified statement is the flat-branch eigenvalue identity

$$H_{4,4}^{\text{exc}}(k)\psi(k) = \Delta q_4 \psi(k).$$

This distinction does not change q_4, A_4, B_4, the bandwidth, or the global extrema.

2.4 Appendix C. Notation and coefficient conversion registry

To prevent the documentary collision that caused earlier status errors, use:

- q_N: momentum-independent fourth-order branch shift;
- $(\alpha_N^{\text{pen}}, \beta_N^{\text{pen}})$: coefficients in the older two-invariant numerator pencil;
- $(A_N^{\text{shp}}, B_N^{\text{shp}}, C_N^{\text{shp}}, D_N^{\text{shp}})$: coefficients in the generic four-shape basis.

The exact conversion is

$$A_N^{\text{shp}} = \frac{\alpha_N^{\text{pen}}}{4}, \qquad B_N^{\text{shp}} = 0, \qquad C_N^{\text{shp}} = \frac{\beta_N^{\text{pen}} - 2\alpha_N^{\text{pen}}}{16}, \qquad D_N^{\text{shp}} = 0$$

for every currently solved physical rank.

Rank	q_N	α_N^{pen}	β_N^{pen}	Four-shape status
$SU(3)$	exact rational	5/12	exact rational; stable expression plus $-25/64$ anomaly	exact; $B_3^{\text{shp}} = D_3^{\text{shp}} = 0$
$SU(4)$	exact rational, including determinant shift	32/675	exact rational; equals stable expression	exact; $B_4^{\text{shp}} = D_4^{\text{shp}} = 0$
$SU(5)$	finite-rank certificate; no determinant correction	1/108	certified to equal stable expression	exact Layer-II rank closure; $B_5^{\text{shp}} = D_5^{\text{shp}} = 0$
$SU(6)$	finite-rank certificate with $\Delta q_6 = 6/343$	64/25725	certified to equal stable expression	exact Layer-II rank closure; $B_6^{\text{shp}} = D_6^{\text{shp}} = 0$
$SU(N \geq 7)$	exact symbolic stable-rank family	$640/[N(N^2-1)^3]$	$P_{402}(N)/D_{409}(N)$	exact family; $B_N^{\text{shp}} = D_N^{\text{shp}} = 0$

2.5 Appendix D. Canonical source and status registry — v1.4

Source	Role	Authority
`MASTER THEORY Gauge Constrained Spectral Geometry 2026-08-01.md`	five-layer narrative spine	synthesis, not sole status authority
`UPDATED_FOURTH_ORDER_HODGE_THEORY_2026-08-01.md`	Hodge-duality, four-shape obstruction space, coefficient conversion	theorem package
`SU3_Compact_Remainder_Proof.docx`	rigorous $O(\beta^{-1})$ compact-gap remainder	analytic proof insertion

Source	Role	Authority
`verify su3 proof.py`	algebra/moment/final-assembly checks	partial executable verification
`SU4_HYBRID_COMPLETE_THEOREM_V2 (2).md`	exact $SU(4)$ exceptional completion	theorem statement
`SU4_HYBRID_COMPLETE_CERTIFICATE_V2.json`	exact gates, coefficients, hashes, residuals	certificate
`SU4_EXCEPTIONAL_TOPOLOGY_AND_WORD_LEDGER_V2.json`	96 topology records and 76 word corrections	detailed ledger
`PAPER homological flat bands.md` (2026-08-08)	Layer-II rank synthesis: universal axial law, $SU(5)/SU(6)$ closure, exceptional-rank partition, $SU(3)$ second-pencil anomaly	rank/status authority relative to accepted certificates
`Pasted markdown.md` (2026-08-08 independent SIMULATIONS audit)	independent re-derivation and proof-chain audit of the $SU(3)$ strong-coupling simulation corpus	proof-status authority for $SU(3)$ $O(y^4)$ certificate completeness

2.5.1 v1.4 status correction

The older statement that $SU(5)$ and $SU(6)$ are unresolved is superseded by the 2026-08-08 Layer-II authority. The universal axial coefficient is exact for every integer $N \geq 3$. The generic four-shape ambient space remains four-dimensional, while the physical contraction satisfies $B_N^{\text{shp}} = D_N^{\text{shp}} = 0$ across the resolved rank partition. The special $SU(3)$ correction $\Delta\beta_3^{\text{pen}} = -25/64$ prevents a universal quotient-scalarity claim; such a claim may only be considered for $N \geq 4$.